

Hot Lithography – New Possibilities in Polymer 3D Printing

A newly developed stereolithography-based additive manufacturing technology combines good material properties with outstanding manufacturing precision

Markus Pfaffinger

Additive manufacturing as an industrial production tool of polymer parts requires outstanding precision and surface quality of the printed objects in combination with good mechanical and thermal material properties. For this, current technologies often provide only insufficient solutions. Cubicure developed the hot lithography technology which allows the processing of high-performance photopolymers in outstanding precision.

Among different strategies and technologies of 3D printing a physical three-dimensional object out of powder, liquid, or solid material, stereolithography (SLA) is the method of choice with respect to highest possible resolution and surface quality of the final printed parts. SLA processes are based on photopolymerization of a light sensitive resin. Here, a UV laser is used to expose a light sensitive resin and selectively cures it in a layer-by-layer approach. Since its invention and commercialization in the early 1980s, the material properties of the cured resins are the biggest challenge in SLA technologies. Conventional low viscosity photopolymers for SLA are very rigid, but also brittle or soft with a low glass transition temperature. Hence, they are subjected only to a limited extent for technical applications [1, 2].

On the other hand, there are technologies like selective laser sintering (SLS) or fused deposition modeling (FDM) which deliver reasonable material properties but rather poor surface qualities of the printed parts [1].

Consequently, there currently is no technology in AM of polymers that combines both good material properties and good surfaces. This is called the dilemma of polymer 3D printing (Fig. 2).

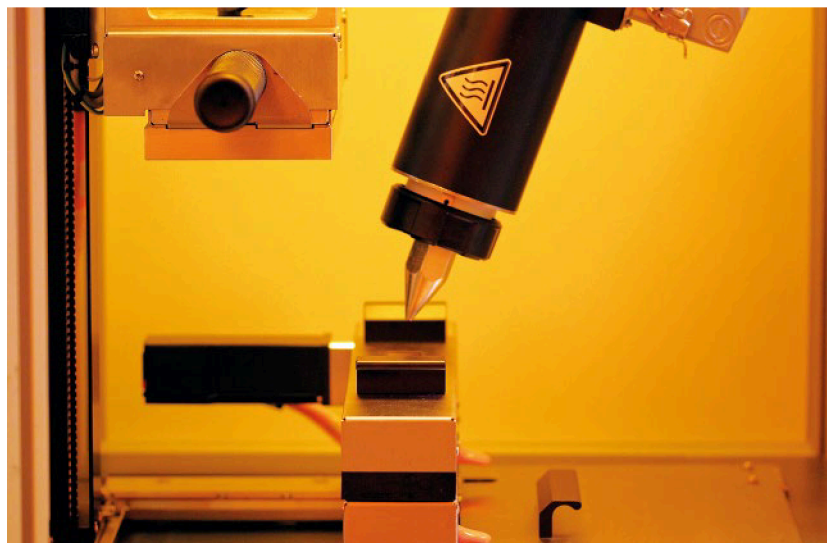


Fig. 1 Process chamber of Caligma 200 (all images source: Cubicure)

The aim of Cubicure, a spin-off of the TU Wien, is the additive manufacturing of polymer parts with outstanding material properties, such as high impact strength and high heat deflection temperature while, at the same time, having the highest possible surface quality. Thus, printing fully functional polymer parts for industrial applications and objects of daily life comparable with parts produced by injection molding.

To achieve this, an SLA-based technology needs to be used to meet the geometric requirements (i.e. resolution and surface quality). In addition, it is necessary to improve the mechanical and thermal properties of photocurable resins towards the property profile of technical thermoplasts, such as polypropylene (PP) or acrylonitrile butadiene styrene (ABS).

This is possible through newly developed photopolymers in addition with a new and innovative 3D printing process, the hot lithography.

Cubicure started its material research in 2015. Already during theo-

retical considerations and approaches about suitable chemical systems and compounds for those photopolymer systems, it was evident that the resulting substances would be highly viscous at room temperature. To achieve thermoplastic-like properties, rather large resin molecules with a high molecular weight are favored. This results in highly viscous and sticky raw materials which cannot be processed in commercial SLA machines until now.

Out of this, in parallel to the material development, Cubicure started developing its own special 3D printing concept which is applicable for the processing of

Meet us at
Formnext
Hall 3.1
Booth G59

	GEOMETRY	MATERIAL	ECONOMICS
SELECTIVE LASER SINTERING	~	✓	✗
FUSED DEPOSITION MODELING	✗	✓	~
INKJET PHOTOPOLYMERS	✓	✗	✓
STEREOLITHOGRAPHY	✓	✗	✓

Fig. 2 Dilemma of polymer 3D-printing

those poorly flowable material systems. Due to a strong temperature dependency of a fluid's viscosity, a heated 3D printing process was implemented. The multiple patented technical solution of this process was published under the name hot lithography. This technology enables the usage of high-molecular chemical substances for a high precision 3D printing process for the first time and consequently, establishes parts with unexpected material quality.

Hot lithography technology

Conventional SLA systems operate at room temperature. The flowable low viscous resin is supplied in a tank bath where on the resin's surface the laser scanning, hence, the polymerization and the three-dimensional shaping happens. The low viscosity of the resin is due to their very low molecule size meaning low molecular weight. This leads to highly crosslinked networks in the printed parts, causing brittleness and poor dimensional stability under heat.

In contrast, the hot lithography technology's core is a specially developed heated coating mechanism which is capable of processing resins and pastes of the highest viscosity at operating temperatures up to 120 °C. A heated transparent plate moves below the coating unit and carries thin layers of the photosensitive material into the processing area (Fig. 3). The laser is placed below the carrier plate and selectively cures the resin through the transparent plate in a bottom-up layer-by-layer approach. The unique setup of the material carrier plate guarantees a constant operating

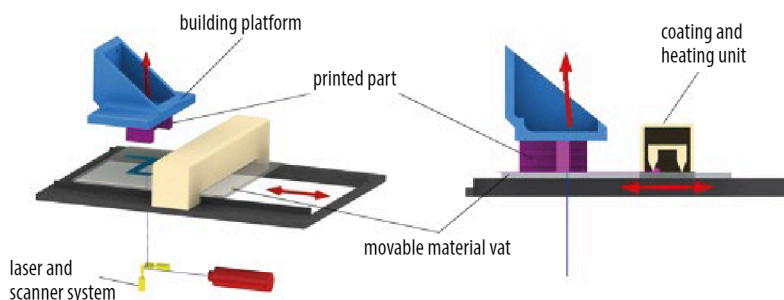


Fig. 3 Process scheme of the hot lithography technology

temperature of the photopolymer without overheating it. Temperature peaks and local overheating of the resin must be strictly avoided. Elevated temperatures do not only affect the viscosity of a resin, but also its stability and reactivity. Hence, the temperature of all process elements need to be precisely controlled to avoid unintended polymerization and consequently, the degradation of the material. A well-defined process temperature makes it possible to run a safe and stable printing process for days. After each curing cycle, the remaining uncured photopolymer on the material carrier plate can be used in the following cycle. The newly developed hot coating technology got patented.

With this combination of material and process it is possible to produce photopolymer-based plastic components which allow applications in industrial and technical environments. Hot lithography enables the additive processing of technically relevant plastics of highest precision for the first time.

Caligma 200 – the first commercial hot lithography 3D printer

Cubicure's first hot lithography 3D printer, named Caligma 200 (Fig. 4), had its market launch in June 2017. During development the machine engineering was focused on the combination of flexibility of usable materials and the greatest possible process safety.

The Caligma 200 can process any photocurable resin almost independently of its viscosity. However, it is optimized to process highly viscous resins and pastes at temperatures up to 120 °C. The maximum processable viscosity is about 20 Pa·s at operating temperature.

The Caligma 200 production plant incorporates a high-end laser light en-

gine for the finest manufacturing precision. On customer request, the machine can be configured with a laser wavelength in the visible range (405 nm, Caligma 200) or in the UV range (375 nm, Caligma 200 UV). The used diode lasers are from premium segments of industrial suppliers and guarantee highest beam quality ($M^2 < 1.15$) as well as high power constancy over a long lifetime. The laser spot size is also adapted during machine construction and can be selected in the range between 10 and 100 μm. Hereby, precision trades for process economics. Cubicure's standard setup features a 25 μm laser focus diameter. With this high precision laser system also, a specific surface texturing of parts is possible (Fig. 5).

The printer comes with pre-defined process parameters for Cubicure's standard materials. This ensures easy operating through the customer and reproducible printing results.

Latest PLC control technology ensures constant reliable printing conditions and process stability. A long-term operation of several days without a break is not a problem.

Low power consumption of about 400 W (at 230 or 115 V) during printing and an activated carbon filter enable an operation in office environment.

Cubicure provides its resins in 300 ml cartridges which can be loaded and exchanged easily in a heated cartridge holder inside the processing chamber of the Caligma 200 (Fig. 1). The machine monitors the material filling level inside the coating unit as well as the loading status of the cartridge. This controlled and automatic dispensing system adds material to the coating unit if necessary to ensure a constant filling level in the coater. If a cartridge is empty the user gets notified. For a stable non-stop printing process, the change of car-

Company

Cubicure

Vienna, Austria

Cubicure GmbH was founded in March 2015 as a spin-off of TU Wien. The company deals with the additive manufacturing of high performance polymers for industrial applications. Since 2017, Cubicure offers a specially developed and patented 3D-printing plant, Caligma 200, and associated materials in the market. The aim of Cubicure GmbH is the creation of products of highest quality for an industrial production environment.

www.cubicure.com



Fig. 4 Hot lithography-based production site Caligma 200 of Cubicure

tridges is possible also during a printing job.

The building envelop of 200 mm × 100 mm enables part and series production up to a height of 300 mm at each printing cycle with an adjustable layer height from 10 to 100 µm. Depending on the size of parts, this platform dimension allows the production of functional prototypes with a low lot size which is mainly used to speed up product development processes and consequently reducing the time to market, or especially for small parts, also a series production of several hundreds or even thousands of end-product parts.

The target market of the Caligma 200 are industrial customers from the high-tech sector as well as research organizations. Furthermore, it aims at additive manufacturing service providers who we are interested in enlarging their portfolios of 3D printing technologies.

Current customers are out of diverse industries, namely engineering, supplier's industry, electronics, energy and medical technology.

Innovative photopolymers for hot lithography

Shape on its own does not fulfill function. In the world of industrial parts and high-quality user products, the material is crucial. The demands are as manifold as the production environment itself.

Cubicure is continuously developing new materials to expand the fields of applications and customer industries of its hot lithography technology. In photopolymerization-based processes during 3D printing not only the geometry gets generated but also the material properties become determined. Therefore, the understanding of the underlying structure-property relations is crucial for improving properties of materials.

R&D focus of Cubicure is the toughening of photopolymers and gaining a high dimensional stability of the printed parts under heat.

This can be achieved e.g. by using high molecular resins which are highly viscous or even solid at room temperature and lead to a reduced network density in the cured part. Thereby, a cross-linked thermoset polymer gets properties inspired by a thermoplastic material such as high strength while at the same time high elongation at break.

Beside the photopolymers Cubicure Evolution (for functional prototyping applications) and Cubicure Precision (for high precision and micro parts), the company offers the possibility to customize material properties or to develop new photopolymer formulations. The company represents the entire value chain from material development and production to stereolithographic processing. Thus, application specific requirements to 3D printed polymer parts can be mastered.

250 °C – not a problem!

At the Formnext fair, Cubicure launches a new innovative photopolymer. It is very strong and durable, for operational temperatures of 250 °C and higher. This opens additional fields of applications like mold tooling and temperature-loaded machine parts for the hot lithography technology.

Among the numerous advantages of this new material, it is most importantly now possible to change the base chemistry for lithography-based additive manufacturing. Other than traditional acrylate and meth-acrylate based polymer backbones, the new chemical substances bring unique performance to additive manufacturing which for decades suffered from its material limitations. Using this latest material, the user is now able to directly manufacture



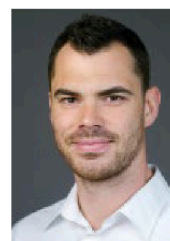
Fig. 5 3D printed functional prototype with textured surface (material: Cubicure Evolution)

ready-to-use parts with tremendous accuracy, surface quality, and outstanding material properties.

DOI: 10.1002/latj.201800024

- [1] S. C. Ligon, R. Liska, J. Stampfl, M. Gurr, R. Mühlaupt: Polymers for 3D Printing and Customized Additive Manufacturing, *Chem. Rev.* 117 (2017) 10212–10290. doi:10.1021/acs.chemrev.7b00074.
- [2] D. Kazmer: 28 - Three-Dimensional Printing of Plastics A2 – Kutz, Myer, in: *Appl. Plast. Eng. Handb. Second Ed.*, William Andrew Publishing, 2017: pp. 617–634. doi:10.1016/B978-0-323-39040-8.00029-8.

Author



Markus Pfaffinger

studied biomedical engineering at the UAS Technikum Wien and subsequently at TU Wien. In 2017, he received a PhD for his work in the Christian Doppler Laboratory for photopolymers

in digital and restorative dentistry in the working group of Prof. Stampfl at TU Wien. At Cubicure, he is responsible for business development since 2017.

Dr. Markus Pfaffinger, Cubicure GmbH, Gutheil-Schoder-Gasse 17, 1230 Vienna, Austria, Tel: +43 1 5810439 10; e-mail: info@cubicure.com, www.cubicure.com